

Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

Achim BAUMGÄRTNER¹, Erik WEIGAND¹,
Gian Bruno TRUÖL^{1,2}, Mirko SONNTAG^{1,2}

¹University of Applied Sciences Esslingen, Esslingen University, Germany

²Fraunhofer Application Center Keim, Esslingen, Germany

acbait01@hs-esslingen.de, erwewi01@hs-esslingen.de,
gian.ledl-truol@hs-esslingen.de, mirko.sonntag@hs-esslingen.de

Submitted: 01 August 2026 | Revised: 25 March 2026 | Accepted: 01 August 2026

Abstract. This papers goal is the investigation on how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

BEAM	Behavior, Energy, Autonomy, and Mobility
EV	Electric Vehicle
MATSim	Multi-Agent Transport Simulation
POI	Point of Interest
SoC	State of Charge
SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
V2G	Vehicle-to-Grid
OSM	OpenStreetMap
CS	Charging Station

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension: vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), turning charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial

and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From land-use Point of Interests (POIs)—residential buildings, offices, and amenities—we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface, which is inspired by SUMO’s web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns; a detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy

exchange. This means that the spatial distribution of charging stations can have a significant impact on both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore crucial for effective infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the aforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.

- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load; bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose: SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach than SUMO, is a mesoscopic traffic simulation model that uses a network of nodes and links to represent the road network, and simulates traffic flow based on traffic demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent

Transport Simulation (MATSim) is an example of a mesoscopic traffic simulation model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12].

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the electrical network [14]. While a reasonable amount of studies have been conducted on the siting of charging stations, there are only few that consider working with large-scale electric vehicle charging infrastructure [15].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

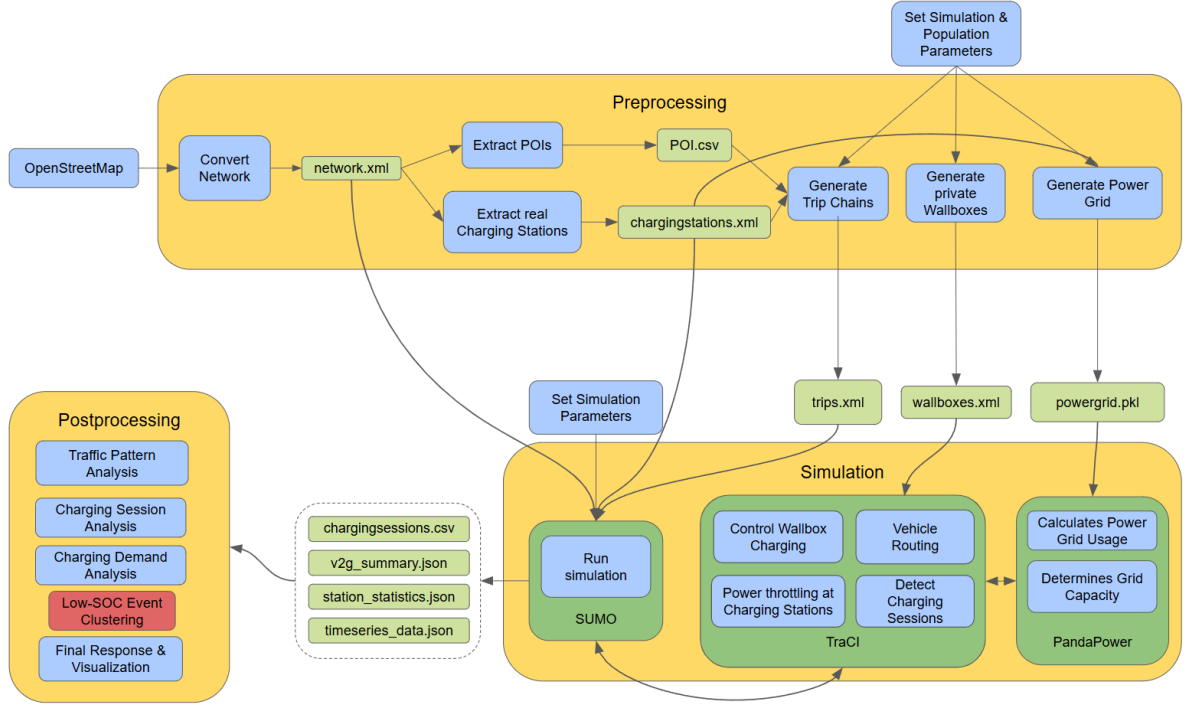


Figure 1: Complete Workflow Pipeline: From OSM data acquisition to scenario generation, execution and manipulation to analysis and visualization.

3. Methodology

SUMO is used as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two siting strategies – *Saturate-and-Prune* and *Clustering* – are implemented and detailed in section 3.2.

3.1 System Overview

Figure 1 illustrates the complete workflow pipeline. The framework is organized into four consecutive stages: (i) data acquisition, (ii) preprocessing and scenario generation, (iii) coupled traffic-grid simulation, and (iv) analysis and visualization. Each stage produces intermediate artifacts that are consumed by the subsequent stage, so that the entire process is reproducible from a single set of user-defined parameters.

Data acquisition. The pipeline retrieves information like the road network, electrical grid topology, and candidate POIs from OSM.

The road network is converted into a SUMO-compatible format using the `netconvert` tool, while grid features and POIs are persisted for later preprocessing. This ensures that the simulation has accurate and relevant data to work with, allowing for realistic modeling of traffic flow and charging behavior in the region of interest.

Preprocessing and scenario generation.

The `chargingstations.xml` file contains both, existing and candidate stations produced by the placement algorithm, and is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. Trip chains map individual agents to a sequence of activities and destinations, from which their charging demand and low-State of Charge (SoC) behavior emerge. In parallel, the private-wallbox generator assigns home-charging devices to a share of agents following a statistical distribution, and the power-grid generator couples the charging stations and wallboxes to their nearest buses to form the synthetic distribution grid. To finalize the preprocessing, user-set parameters – such as the

EV penetration rate, wallbox share, battery capacities, and grid voltage levels – are applied to the mechanisms of the trip chain, private wallbox, and power grid generation, yielding a fully specified, deterministic scenario.

Coupled traffic-grid simulation. After the preprocessing the files `osm.passenger.trips.xml`, `private_wallboxes.xml`, and `power_grid.pkl` will then be used by SUMO to simulate traffic flow and charging behavior. TraCI, SUMO and PandaPower are the key components of the simulation, which is executed in a time-stepped manner. TraCI allows for real-time control of the simulation, enabling the dynamic adjustment of traffic flow and charging behavior based on an Application Programming Interface (API) via Python. SUMO simulates traffic on a microscopic level (i.e., individual vehicles and their interactions), while PandaPower evaluates the impact of charging behavior on the synthetic distribution grid. While SUMO and TraCI handle the traffic simulation, SUMO and PandaPower do not communicate directly. Instead, TraCI acts as a bridge between the two, allowing for the exchange of information and control signals. This enables the simulation to capture the complex interactions between traffic flow and grid behavior, providing the ability for SUMO to incorporate grid constraints and for PandaPower to calculate the impact of charging behavior on the grid.

Analysis and visualization. The simulation produces a variety of output files, including `station_statistics.json`, `v2g_summary.json`, `charts_data.json`, `charging_sessions.csv` and `model_log_data.csv`. These persisted outputs are analyzed and visualized to evaluate the placement algorithm along metrics such as charging sessions, net energy flow over time, peak grid power, and individual charging station utilization (section 3.3). The results are rendered as plots and maps in the web interface, and inform future iterations of the

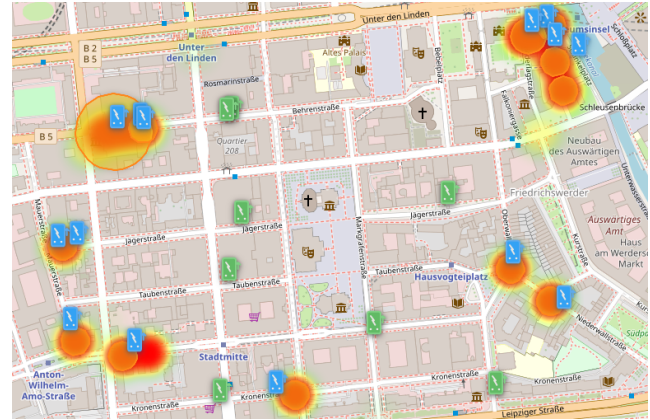


Figure 2: Resulting visualization with generated (blue) and extracted (green) CS for example run.

placement algorithm. Figure 2 shows an example visualization with generated (blue) and extracted (green) charging stations for a single run: the heatmap and orange circles indicate approximate areas for charging station placement, while the blue and green dots represent the actual locations of the generated and extracted charging stations, respectively.

3.2 Charging Station Placement Algorithm

Two siting strategies are implemented to compare different placement policies:

- **Saturate-and-Prune:** A naive iterative approach where candidate stations are placed on all viable road segments, ensuring maximum coverage. After each trial execution, charging stations with less than a 5% utilization rate are removed, and the simulation is rerun until all remaining stations exceed this threshold. A 5% threshold was chosen to ensure that stations are neither underutilized nor overburdened, while also allowing for a sufficient number of stations to be placed. This approach is designed to maximize accessibility and coverage, but may result in a higher number of stations being placed than necessary, leading to potential underutilization of some stations.
- **Clustering:** A heuristic that prioritizes locations with high traffic flow, potential

traffic congestion, and consideration for vehicles that enter a state of low SoC. Placement follows the traffic flow, aiming to maximize accessibility where vehicles are most likely to need charging while reducing detours.

3.3 Evaluation Metrics

Mobility accessibility. CS-placement is evaluated based on traffic flow, charging behavior, energy consumption, and SoC. The gathered data is used to calculate the accessibility of the charging stations, which is defined as the percentage of agents that are able to access a charging station within a certain distance or time threshold. Distance and time are key factors in determining the accessibility of charging stations, as they directly impact the convenience and feasibility of using the stations for charging. Distance is an honest measure of how far an agent must travel to reach a charging station, while time takes into account factors such as traffic conditions, congestion, timely availability of charging stations, time and power (kWh) required to charge.

Grid impact. During the simulation, PandaPower solves an AC power flow on the synthetic distribution grid at every step, evaluating bus voltages and the loading of lines and transformers as constraints that bound the power allocated to each station. The grid impact reported per run is the aggregate demand and the peak grid power relative to the available grid capacity, which captures the stress the charging activity imposes on the grid without exceeding its voltage and loading limits.

Aggregation. The evaluation metrics are aggregated to provide a comprehensive assessment of the charging station placement algorithm. This includes combining mobility accessibility and grid impact metrics to evaluate the overall performance of the charging station network. The aggregation process involves analyzing the trade-offs between accessibility and grid impact,

allowing for a balanced evaluation of the placement algorithm's effectiveness.

4. Experiments

4.1 Experimental Setup

4.2 Results

5. Discussion

5.1 Limitations

5.2 Interpretation of Results

5.3 Practical Example

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

Acknowledgments

The authors thank the Fraunhofer Center of Application KEIM for supporting this research work.

References

- [1] International Energy Agency (IEA), *Global EV Outlook 2023: Catching up with Climate Ambitions*. Paris, France: IEA, 2023. [Online]. Available: <https://www.iea.org/reports/global-ev-outlook-2023>

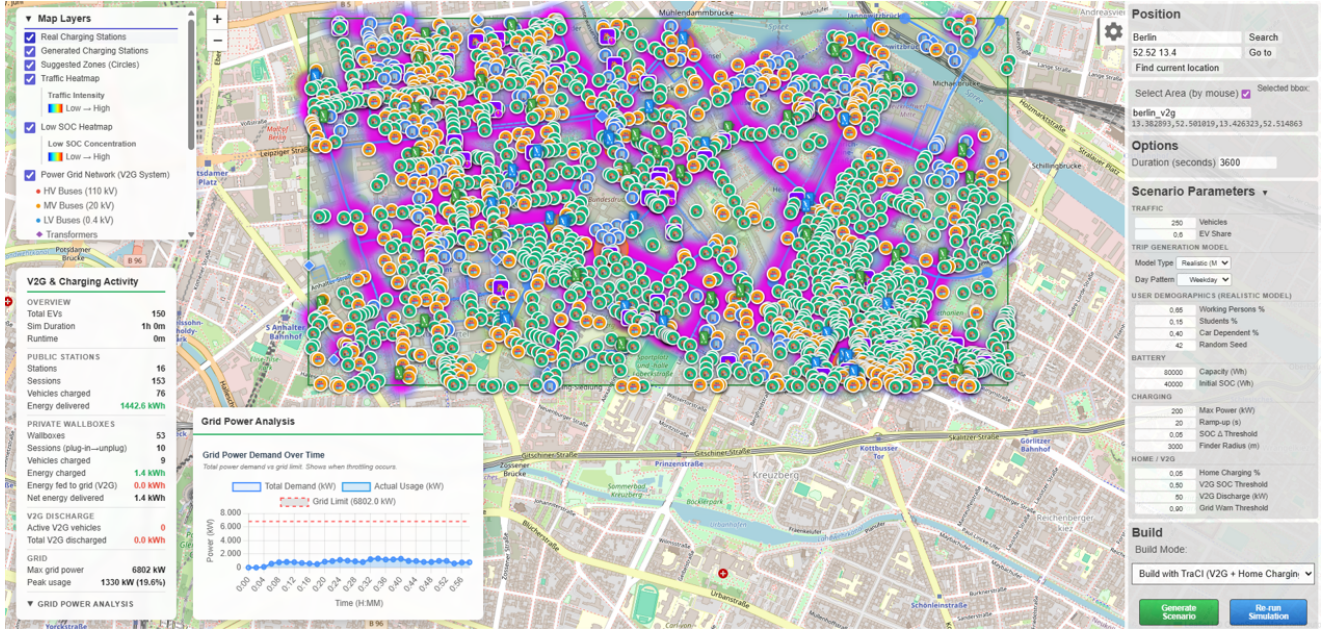


Figure 3: Example run of the workflow: the web interface with all data layers enabled.

- [2] W. Kempton and J. Tomic, “Vehicle-to-grid power fundamentals: Calculating capacity and net revenue,” *Journal of Power Sources*, vol. 144, no. 1, pp. 268–279, 2005.
- [3] K. Clement-Nyns, E. Haesen, and J. Driesen, “The impact of charging plug-in hybrid electric vehicles on a residential distribution grid,” *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 371–380, 2010.
- [4] F. He, D. Wu, Y. Yin, and Y. Guan, “Optimal deployment of public charging stations for plug-in hybrid electric vehicles,” *Transportation Research Part B: Methodological*, vol. 47, pp. 87–101, 2013.
- [5] “A review on electric vehicle charging station planning: Infrastructure placement, sizing, upgrades, and uncertainties,” *Journal of Energy Storage*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352152X25040381>
- [6] “Data-driven optimization of EV charging station placement using causal discovery,” *arXiv preprint arXiv:2503.17055*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.17055>
- [7] T. Kurczveil, P. Á. López, and E. Schnieder, “Implementation of an energy model and a charging infrastructure in SUMO,” in *Simulation of Urban Mobility (SUMO)*, https://www.researchgate.net/publication/276063158_Implementation_of_an_Energy_Model_and_a_Charging_Infrastructure_in_SUMO,
- [8] “Assignment approach for electric vehicle charging using traffic data collected by SUMO,” *World Electric Vehicle Journal*, vol. 14, no. 2, art. 40, 2023. [Online]. Available: <https://www.mdpi.com/2032-6653/14/2/40>
- [9] “Vehicle-to-grid enabled charging infrastructure planning and operations considering demand uncertainties,” *Transportation Research Part D: Transport and Environment*, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1361920923003152>
- [10] “Assessing EV charging impacts on power distribution systems: A unified co-simulation framework,” *arXiv preprint arXiv:2505.21773*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.21773>

- [11] A. Horni, K. Nagel, and K. W. Axhausen, Eds., *The Multi-Agent Transport Simulation MATSim*. London, U.K.: Ubiquity Press, 2016. [Online]. Available: <https://doi.org/10.5334/baw>
- [12] C. Sheppard, R. Waraich, A. Campbell, A. Gopal, and A. Pozdnukhov, “Modeling plug-in electric vehicle charging demand with BEAM: The framework for behavior, energy, autonomy, and mobility,” Lawrence Berkeley National Laboratory, Berkeley, CA, USA, Tech. Rep. LBNL-2001018, 2017. [Online]. Available: <https://www.osti.gov/biblio/1398472>
- [13] R. Pagany, L. Ramirez Camargo, and W. Dorner, “A review of spatial localization methodologies for the electric vehicle charging infrastructure,” *International Journal of Sustainable Transportation*, vol. 13, no. 6, pp. 433–449, 2019. [Online]. Available: <https://doi.org/10.1080/15568318.2018.1481243>
- [14] F. Ahmad, A. Iqbal, I. Ashraf, M. Marzband, and I. Khan, “Optimal location of electric vehicle charging station and its impact on distribution network: A review,” *Energy Reports*, vol. 8, pp. 2314–2333, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352484722001809>
- [15] “Large-scale optimization of electric vehicle charging infrastructure,” in *Proc. 32nd ACM Int. Conf. on Advances in Geographic Information Systems (SIGSPATIAL '24)*, 2024. [Online]. Available: <https://dl.acm.org/doi/10.1145/3678717.3700830>
- [16] V. Lazari and A. Chassiakos, “Multi-objective optimization of electric vehicle charging station deployment using genetic algorithms,” *Applied Sciences*, vol. 13, no. 8, art. 4867, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/8/4867>